

Iniziato martedì, 17 giugno 2025, 09:48**Stato** Completato**Terminato** martedì, 17 giugno 2025, 09:58**Tempo impiegato** 9 min. 38 secondi**Domanda 1**

Risposta corretta

Punteggio max.: 2,00

Let f, g be the functions defined as $f(n) = 10^4 2^{3n} \log n$ and $g(n) = 10^{-5} 2^{4n} \log n$.

Scegli una o più alternative:

- ☐ $f \in \Theta(g)$
- ☒ $f \in O(g)$ ✓
- ☐ $f \in \Omega(g)$

Domanda 2

Risposta corretta

Punteggio max.: 2,00

Which of the following sets can be easily encoded as binary strings?

Scegli una o più alternative:

- ☒ Strings on an alphabet of size 10^3 ✓
- ☒ Finite sequences, each element in $\{0, 1\}^*$ ✓
- ☒ Unirected graphs ✓
- ☐ Vectors of complex numbers

Domanda 3

Risposta corretta

Punteggio max.: 2,00

The problem of deciding whether an undirected graph G has an Hamiltonian cycle is:

Scegli una o più alternative:

- ☐ Undecidable, and **NP**-hard.
- ☒ Decidable in polynomial time, if $P = NP$ ✓
- ☐ Not in the class **NP**
- ☒ In the class **EXP** ✓

Domanda 4

Risposta corretta

Punteggio max.: 2,00

A problem $\square$ is said to be **EXP**-complete if it is in **EXP** and any problem in **EXP** can be reduced, in polynomial time, to $\square$. Suppose a friend of yours comes to you claiming that he found an **EXP**-complete problem which is also in **NP**. What can you say?

Scegli una o più alternative:

- ☒ That your friend is perhaps right, but that, if right, he would have proved $P \neq NP$. ✓
- ☐ That your friend is certainly right, because $P = EXP$.
- ☐ That your friend is perhaps right, because finding such a problem is very easy.
- ☐ That your friend is certainly wrong, because such a problem cannot exist.

Domanda 5

Risposta corretta

Punteggio max.: 2,00

The concept classes of axis-aligned squares in $\mathbb{R}^2$ is:

Scegli una o più alternative:

- ☒ Efficiently PAC learnable, having finite VC-dimension. ✓
- ☒ PAC learnable ✓
- ☐ Not PAC learnable
- ☐ PAC learnable, but not efficiently so, having infinite VC-dimension.

Vai a...

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